Pregătire Examen – LFA 2022

$$L_1 = \{ a^n b^m (bc)^{n\%3} \mid m, n > 0 \}$$

(M_3) $S \to aS_1 \mid bS_{30}$

(M_3+1) $S_1 \to aS_2 \mid bS_{31}$

(M_3+2) $S_2 \to aS \mid bS_{32}$

$S_{30} \to bS_{30} \mid e$

$S_{31} \to bS_{31} \mid bc$

$S_{32} \to bS_{32} \mid bcbc$

GR. pt. $a^n b^m (bc)^{n\%3}$, cu $m>0$, dar $n \geq 0$

Cum modific pt. a tine cont de $n>0$.

(0) $S \to aS_1 \mid \cancel{bS_{30}}$

(M_3+1) $S_1 \to aS_2 \mid bS_{31}$

(M_3+2) $S_2 \to aS_3 \mid bS_{32}$

$S_{30} \to bS_{30} \mid e$

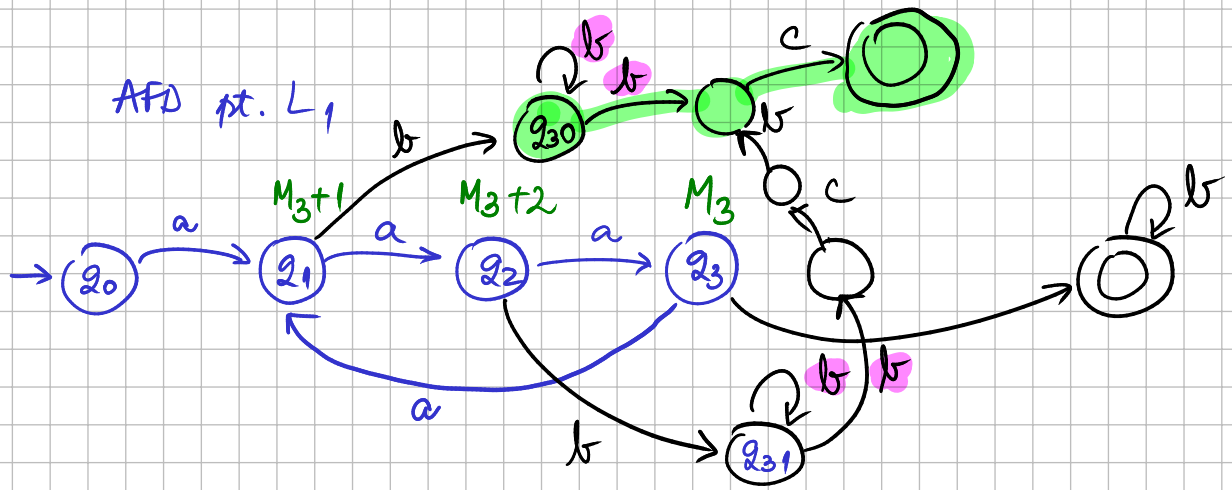
$S_{31} \to bS_{31} \mid bc$

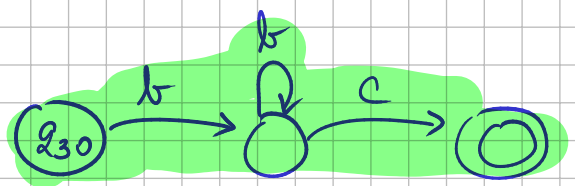
$S_{32} \to bS_{32} \mid bcbc$

(M_3) $S_3 \to aS_1 \mid bS_{30}$

AFN

AFD pt. L_1





$L_2 = \{ w \in \{a,b\}^* \mid$ primul simbol $\neq$ ultimul simbol și $|w|$ este impară $\}$

$S \to aS_1 b \mid bS_1 a$ – G.I.C.

$$\begin{cases} S \to aS_1 \mid bS_2 \\ S_1 \to aaS_1 \mid abS_1 \mid baS_1 \mid bbS_1 \mid ab \mid bb \\ S_2 \to aaS_2 \mid abS_2 \mid baS_2 \mid bbS_2 \mid aa \mid ba \end{cases}$$ – G.R.

ER pt. L_2: $a((a+b)(a+b))^*(ab+bb) + b((a+b)(a+b))^*(aa+ba)$

$L_3 = \{ w \in \{a,b\}^* \mid$ primul simb $\neq$ ultimul simb, $\#(w)$ impară și simbolul din mijloc este a $\}$

$L_3 = L_2 \cap \{ w \in \{a,b\}^* \mid$ simb din mijloc este a $\}$

L_2 ↓ LR

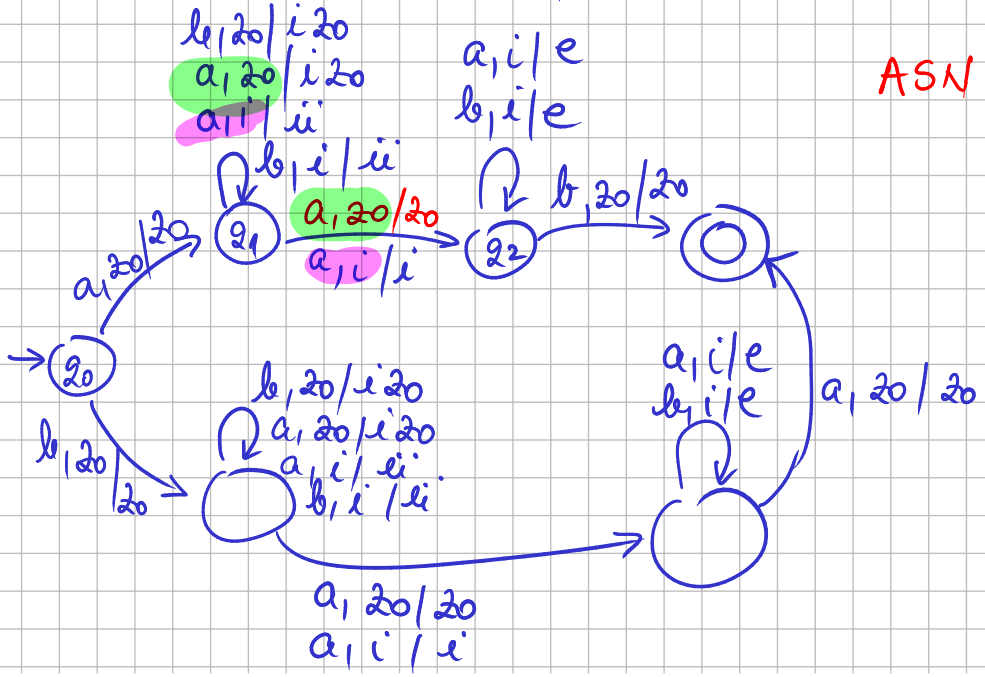
$\{ w \in \{a,b\}^* \mid$ simb din mijloc este a $\}$ ↓ LiC

$uav, u,v \in \{a,b\}^*, |u| = |v|$

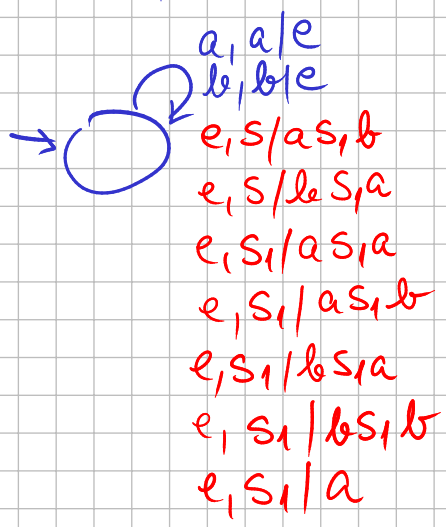
$LR \cap LiC \to LiC$

$$\begin{cases} S \to aS_1 b \mid bS_1 a \\ S_1 \to aS_1 a \mid aS_1 b \mid bS_1 a \mid bS_1 b \mid a \end{cases}$$ G.I.C.

ASN



AS pornind de la G.I.C.



$$\begin{cases} S \to aS_1b \mid bS_1a \\ S_1 \to aS_1a \mid aS_1b \mid bS_1a \mid bS_1b \mid a \end{cases}$$

$L_4 = \{ ww \mid w \in \{a,b\}^*, |w| \% 2 = 0,$ w conține pe poz. pare a și pe poz. impare $b \}$

$w = ba$
$\quad\ 1\ 2$

$|w| = 2k \Rightarrow |ww| = 4k = M_4$

$ww = baba$
$\quad\ 1234$

$L_4 = \{ w \mid w \in \{a,b\}^*,$ w conține pe poz. pare a și pe poz. impare b, $\#(w) \% 4 = 0 \}$

$S \to babaS \mid e$ – G.R.

$\#(w) = 2k+1$

$w = b \Rightarrow ww = bb$

$w = bab \Rightarrow ww = babbbab$

$L_5 = \{ ww^R \mid w \in \{a\}^+, \#(w) \bmod 2 = 1 \}$

$w = w^R$ pt. că $w \in \{a\}^+$

$\#(ww^R) = 2k+1 + 2k+1 = 4k+2 = M_4 + 2$

$S \to aaaaS \mid aa$ – G.R.

$\{a^n b^m \mid n-m=2\}$ $\quad n-m=2 \Rightarrow n=m+2$

$a^n b^m = a^{m+2} b^m = a^2 a^m b^m$

$L_1 L_2 L_3$ unde pt $L_1 - S_1$, $L_2 - S_2$, $L_3 - S_3$

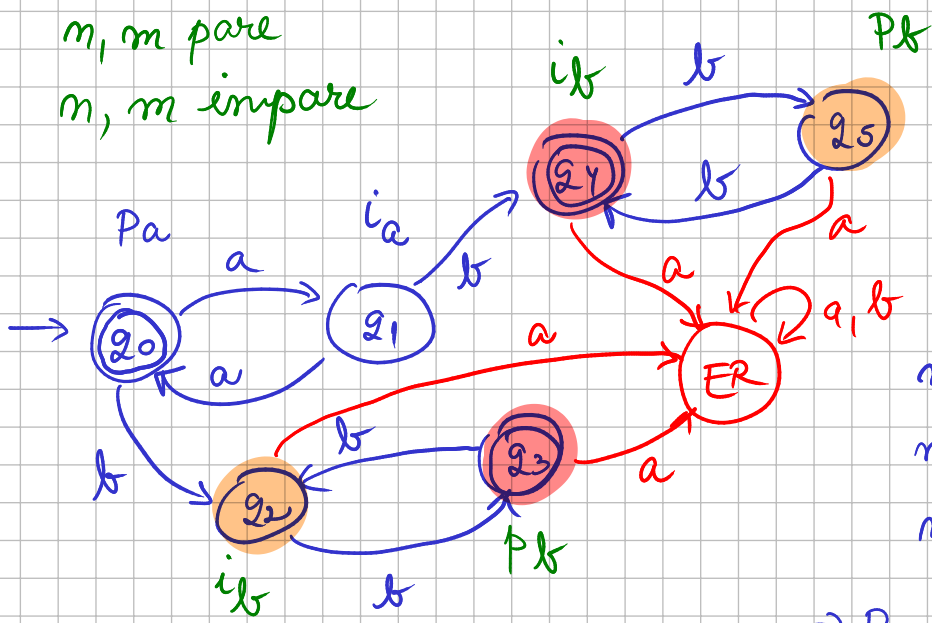
$S \to S_1 S_2 S_3$ – G.I.C

$$\begin{cases} S \to S_1 \\ S_1 \to \ldots \mid S_2 \\ S_2 \to \ldots \mid S_3 \\ S_3 \to \ldots \end{cases}$$

AFD minim pt. $\{a^n b^m \mid n+m = par\}$

n, m pare

n, m impare



$P = \{q_1, q_2, q_5\}$

$F = \{q_3, q_4, q_0\}$

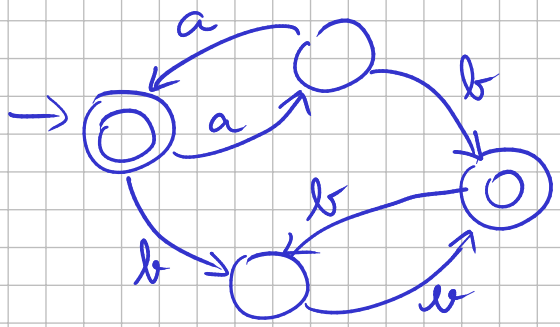
$m(q_1, a) = q_0 \in F$
$m(q_2, a) = ER$
$m(q_5, a) = ER$
$\Rightarrow$

$\Rightarrow P_1 = \{q_1\}$ $P_2 = \{q_2, q_5\}$

$m(q_2, b) = q_3 \in F$
$m(q_5, b) = q_4 \in F$
$\Rightarrow$ $\boxed{q_2 \equiv q_5}$

$m(q_3, a) = ER$
$m(q_4, a) = ER$
$m(q_0, a) = q_1 \in P_1$
$\Rightarrow$ spargem F în $F_1 = \{q_3, q_4\}$ $F_2 = \{q_0\}$

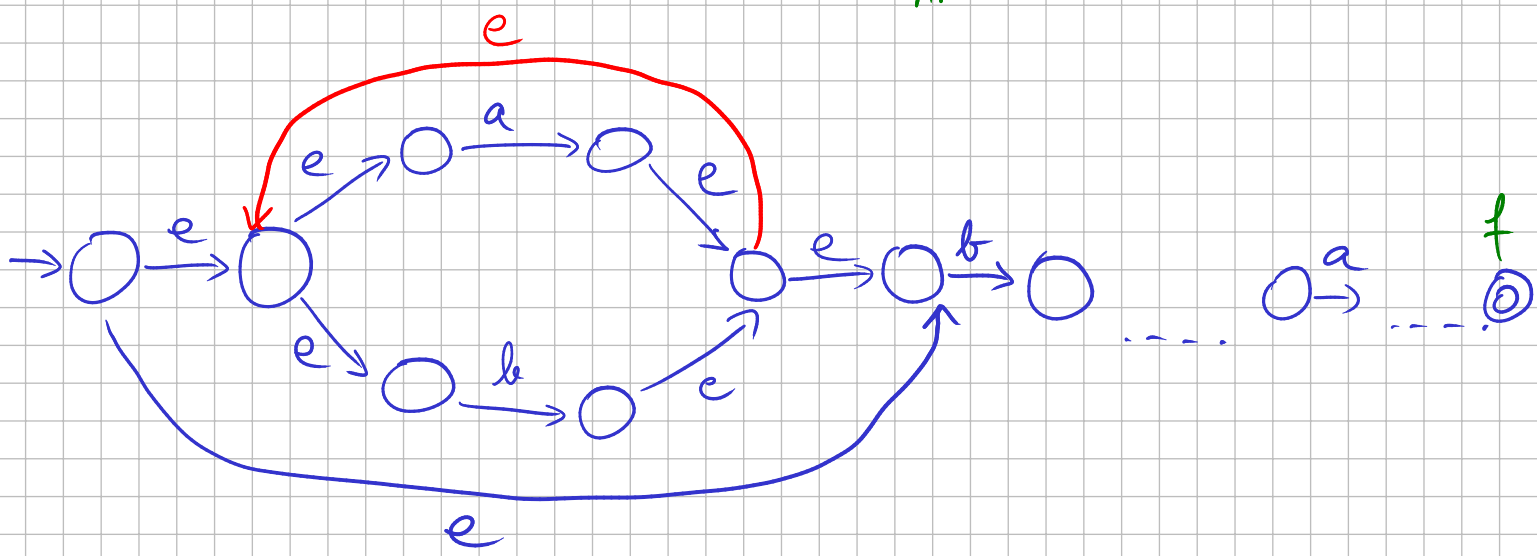
$m(q_3, b) = q_2 \in F_2$
$m(q_4, b) = q_5 \in F_2$
$\Rightarrow \boxed{q_3 \equiv q_4}$



~~AFD~~ minim

ER: $(a+b)^* b (a+b)^* a (a+b)^*$

AFN



$f: \{a,b\}^* \to \{a,b\}^*$, $f(w) = w'$

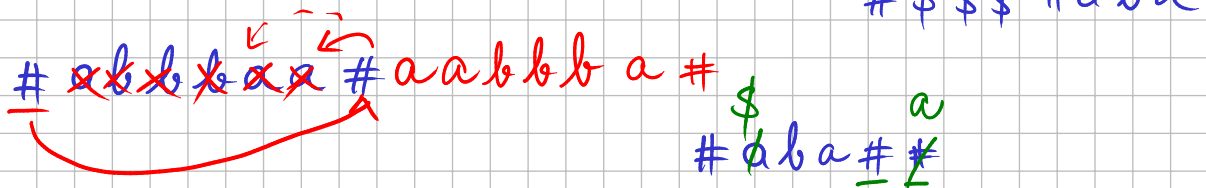
w' = toate aparițiile subșirului ab din w sunt înlocuite cu bb

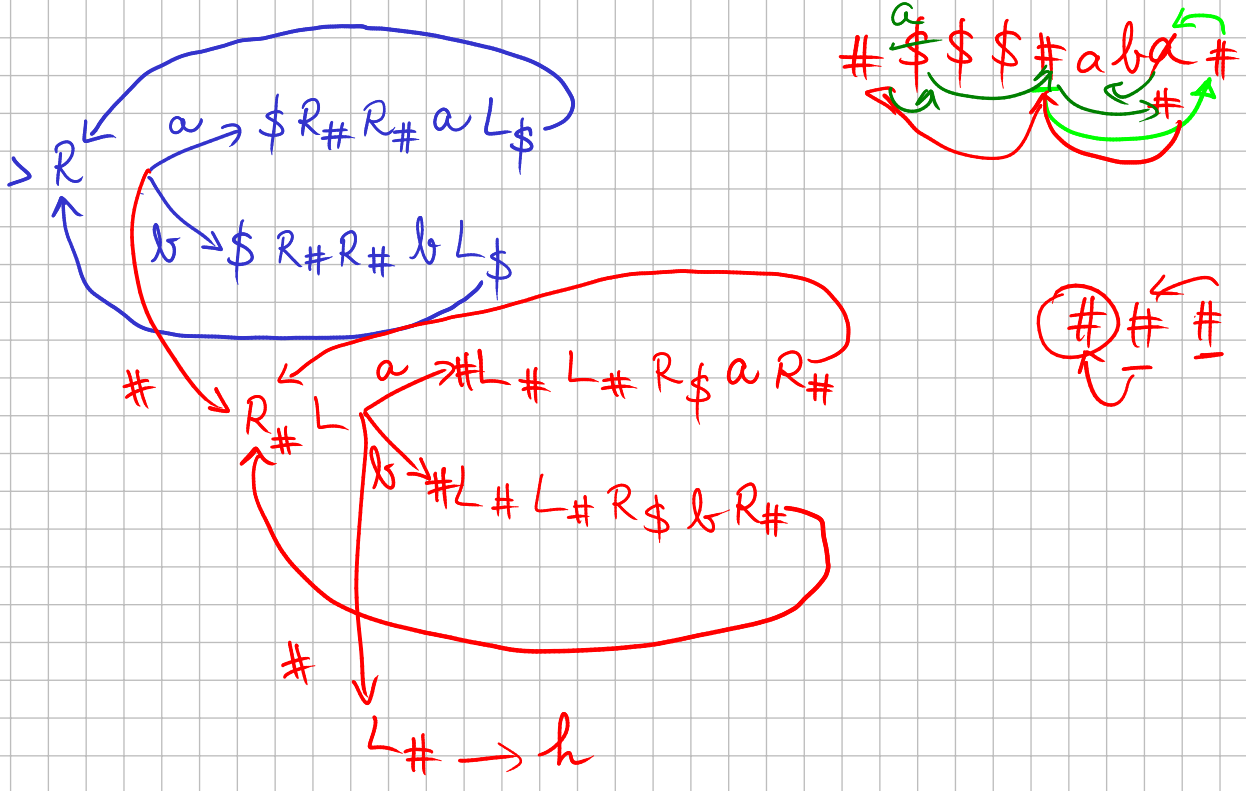
MTC care calculează f.

$\underline{\#}\, a\, b\, a\, a\, b\, b\, a\, b\, a\, \# \longrightarrow \underline{\#}\, b\, b\, b\, b\, b\, b\, b\, b\, a\, \#$

$f: \{a,b\}^* \to \{a,b\}^*$, $f(w) = w^R$

$\#\$\$\#aba$




R
a → $ R# R# a L$
b → $ R# R# b L$
#
R# L
a → #L# L# R$ a R#
b → #L# L# R$ b R#
#
L# → h

a
$ $ $ # a b a
#

#
# #